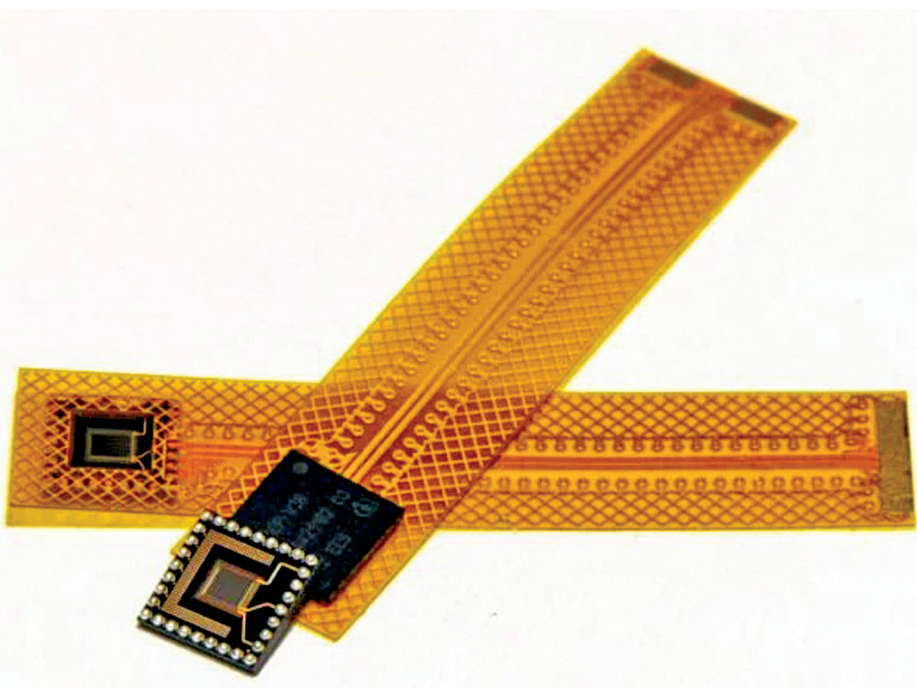




RESEARCH & INNOVATION UPDATES

MEMS-based ASIC that performs much better than piezoelectric materials

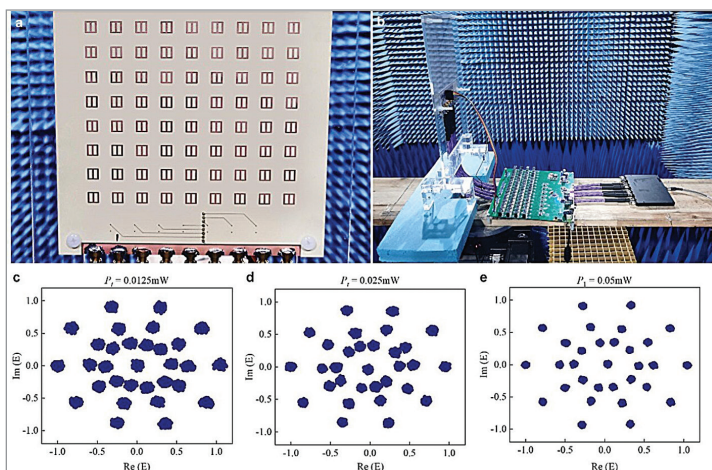


Infineon Technologies AG has developed capacitive micromechanical ultrasonic transducer (CMUT) technology, creating the first MEMS-based one-chip solution. This technology offers a smaller size, better performance, and enhanced functionality. CMUTs generate ultrasonic waves through a deflected semiconductor diaphragm, improving performance over piezoelectric materials. Infineon's MEMS-ASIC integration reduces noise by 20x and amplifies signals by 1000x. Used across industries, CMUTs enable solid-state touch buttons, liquid-level sensing, and advanced medical devices, offering low power, water compatibility, and non-invasive operation.

The ultrasonic technology of Infineon's CMUTsense achieves high signal-to-noise ratio and offers high level of integration (Credit: <https://ee.cdnartwhere.eu/wp-content>)

Programmable metasurface antenna for wireless efficiency

A team from Southeast University, Nanyang Technological University, and other institutions has developed a groundbreaking programmable metasurface to improve wireless communication. Their new design optimises programmable pattern mapping to enhance information retrieval, achieving a 1-bit encoding scheme with spatial harmonics for maximum efficiency. The metasurface, featuring three metallic layers and switchable delay lines, eliminates the need for additional modules, enabling nearly perfect information mapping and paving the way for high-speed, high-capacity networks.



Programmable metasurface antenna (Credit: <https://techxplore.com>)